

Feature Refactor: Convert Bill Calculation into Optional Global Energy Cost Estimator

Context

The website is evolving into a global Electrical Load Assessment platform.

Current features include:

- Electrical Load Calculation
- Daily Energy Consumption
- Monthly Energy Consumption
- Smart Insights
- MCB Recommendation
- Cable Recommendation
- Inverter Recommendation
- Solar Recommendation
- Electrical Health Score
- Energy Saving Analysis
- Professional PDF Reports

The website name is:

Electricity Load Calculator

The primary purpose is electrical planning and decision support.

Electricity bill calculation should no longer be treated as a primary output because:

- Tariffs vary by country
- Tariffs vary by state
- Slab rates differ
- Taxes differ
- Time-of-use pricing exists
- Commercial pricing differs

A fixed bill estimate can be misleading.

The goal is to reposition bill estimation as an optional user-controlled feature.

Objective

Replace:

Estimated Monthly Bill

with:

Optional Energy Cost Estimator

while keeping monthly energy consumption as the primary metric.

UX Strategy

Current report flow:

Load Summary

Total Load

Daily Consumption

Monthly Consumption

Estimated Monthly Bill

Top Consumer

New flow:

Load Summary

Total Load

Daily Consumption

Monthly Consumption

Top Consumer

Smart Insights

Health Score

Energy Cost Estimator (Optional)

Core Principle

Consumption is universal.

Cost is local.

The report should focus on universal metrics first.

Report Changes

Remove Primary Bill Card

Remove the existing prominent card:

Estimated Monthly Bill

from the main summary hierarchy.

Introduce New Section

Title:

Energy Cost Estimator

Subtitle:

Calculate your estimated energy cost using your local electricity rate.

Place this section below:

Health Score

or near the end of the report.

Do not place it above Smart Insights.

User Inputs

Input 1

Electricity Rate

Example placeholders:

0.15

₹8

€0.25

0.30 AUD

Allow numeric entry only.

Unit label:

per kWh

Input 2

Currency Symbol (Optional)

Dropdown:

₹

\$

€

£

AED

AUD

CAD

Other

Default:

Auto-detect from browser locale if available.

Fallback:

\$

Calculation Logic

Use:

Monthly Consumption × Electricity Rate

Example:

753 kWh

×

0.15

=

112.95

Display:

Estimated Monthly Energy Cost

\$112.95

Messaging

Display informational note:

This estimate uses the electricity rate you provide.

Actual utility bills may vary due to taxes, fixed charges, slab pricing, demand charges, and regional tariffs.

Keep wording concise and professional.

PDF Changes

Remove:

Hardcoded bill estimate.

Replace with:

Energy Cost Estimate

Only include if the user has entered a rate.

If no rate is entered:

Do not display a cost estimate in the PDF.

Display only:

Monthly Consumption

Health Score Changes

Do not use cost estimation in Health Score calculations.

Health Score should remain based on:

- Load utilization
- Energy efficiency
- Safety readiness
- Renewable readiness

Cost should not influence health scoring.

Homepage Messaging Updates

Update messaging across the website.

Reduce emphasis on:

Electricity Bill Calculation

Increase emphasis on:

Electrical Load Assessment

Electrical Planning

Energy Consumption Analysis

MCB Selection

Cable Recommendation

Inverter Planning

Solar Planning

Electrical Health Assessment

SEO Strategy

Primary Keywords

Electrical Load Calculator

Load Calculator

Electrical Load Assessment

Home Load Calculator

Connected Load Calculator

Power Consumption Calculator

Secondary Keywords

Energy Cost Estimator

Electricity Cost Calculator

Energy Usage Cost Calculator

The website should be positioned as an electrical planning platform first and a cost estimator second.

Technical Requirements

- Reuse existing monthly consumption value.
 - No backend changes.
 - Fully client-side.
 - Mobile-first design.
 - Maintain existing dark theme and amber branding.
 - Preserve existing report structure.
 - Keep Energy Cost Estimator optional.
 - Ensure users can use the entire platform without entering any tariff information.
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Success Criteria

Users should complete a load calculation and receive:

✓ Load Analysis

✓ Smart Insights

✓ MCB Recommendation

✓ Cable Recommendation

✓ Inverter Recommendation

✓ Solar Recommendation

✓ Electrical Health Score

without needing to enter any electricity tariff.

If users want a cost estimate, they can optionally provide their local rate and instantly calculate their estimated monthly energy cost.

The platform should feel global, engineering-focused, and independent of regional utility pricing models.